

Advanced Matrix Theory (SC 404)

Course Placement: It is an elective course for BTech MnC, ICT and CS program.

Credit Structure: 3-0-0-3

Prerequisite: Linear Algebra (MC215/SC223), Real and Complex analysis (SC107/MC222)

Course Objective: Notions of matrices are of fundamental importance in Mathematics as well as several other areas including Machine learning, Data Science, Computer vision, Networking and Optimization, Computer Science, Signal processing etc. This course aims to develop a strong foundation for matrix analysis, developing essential tools. The emphasis will be on analytic aspects as well as algebraic aspects of Matrix Theory.

Course content: (The selection of topics is variable depending on time constraints.)

1. Review: eigenvalues, eigenvectors, special types of matrices.
2. Schur triangular form, spectral theorem, singular value decomposition, polar decomposition, generalized inverse, least squares solutions.
3. Positive semidefinite and Positive definite matrices: Properties and characterizations
4. Matrix norms, spectral radius formula, relationship between matrix norms and singular values.
5. Perron–Frobenius theory: Matrices with positive and non-negative entries.

Suggested Textbook/references:

- [1] F. Zhang. Matrix theory. Basic results and techniques. Springer, 1999.
[2] R.A. Horn and C.R. Johnson. Matrix analysis. Cambridge University Press, 2012.

Assessment method: In-Sem 1 - 30%, In-Sem 2 - 30%, End-Sem - 40%

Course Outcomes:

After completing the course, the students should be able to:

- ☐ Understand basic concepts related to matrix theory [PO12]
- ☐ Handle abstract ideas of mathematics and learn to write mathematical proofs [PO1, PO2]
- ☐ Understand different types of representations of a matrix [PO4].
- ☐ Develop skills towards solving problems using properties and characterization of special matrices [PO3].

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
X	X	X	X								X

Tentative Lecture Schedule:

Detailed Course Contents		
Topic Name	Content	No. of lectures
Review	Eigenvalues, eigenvectors, special types of matrices, inner product space	2
Matrix decomposition	Schur triangular form, spectral theorem, singular value decomposition, polar decomposition, generalized inverse, least squares solutions.	10
Positive semidefinite matrices	Positive semidefinite and Positive definite matrices: Definition, properties and characterizations	10
Matrix norms	Definition and examples of matrix norms, spectral radius formula, relationship between matrix norms and singular values.	10
Entrywise positive and non-negative matrices	Definition, Properties, Perron's theorem, irreducible matrices, Perron-Frobenius theorem	10